Sumesh Chakkaravarthi Purushothaman

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Education

Northeastern University | *GPA – 3.74* **Expected May 2026**
Master of Professional Studies in Analytics (Concentration in Applied Machine Intelligence) *Boston, MA*

Sri Sairam Engineering College | *GPA – 3.31* **May 2024**
Bachelor of Technology in Artificial Intelligence and Data Science *Chennai, TN*

Technical Skills

Languages: Python, SQL, R, C/C++, Java
Tools: Excel, Tableau, Power BI, PostgreSQL, Redshift, RStudio, Jupyter Notebook, R Markdown, Git, AWS(EC2, S3), Jira
Technical Concepts: Machine Learning, Regression Analysis, Data Cleaning & Modeling, Data Structures, Algorithms, EDA, SHAP, ML Pipelines, Time Series Forecasting, Generative AI, Feature Engineering, Statistical Modeling

Professional Experience

Power Of Patients
Data Science Intern, *Boston, USA* **Apr. 2025 – Jun. 2025**
Tech: Python, NOAA API, Random Forest, Logistic Regression, SHAP, Jupyter Notebook

- Analyzed multi-year health and NOAA weather data to identify correlations between environmental conditions and Sickle Cell Disease (SCD) pain crises, supporting strategic outreach planning.
- Engineered 7-day rolling weather features from NOAA API and merged with clinical datasets by ZIP/date, improving temporal granularity for data modeling and Delivered stakeholder-facing insights through reproducible Python notebooks, aligning visual risk trends with patient geography and seasonality.
- Presented analytical insights through Jupyter-based reports and stakeholder slide decks, translating technical findings into actionable recommendations while maintaining strict NDA compliance.

Cisco AICTE
Data Analyst Intern, *Chennai, India* **Jul. 2023 – Sep. 2023**
Tech: Python, SQL, Excel, Power BI

- Optimized Excel & SQL workflows, improving code efficiency by 20% through data debugging & predictive techniques.
- Enhanced cross-functional tools using Python, reducing errors by 15% and improving performance via data modeling.
- Participated actively in Agile meetings, utilizing strong communication skills and relationship building, while leveraging Power BI to present clear project insights and progress.
- Automated key data processes in Python, cutting manual effort by 30% and applying clustering for efficiency gains.
- Streamlined KPI tracking and reporting through Tableau dashboards and collaborative problem-solving.

Projects

University Explorer AI Chatbot | *Python, PostgreSQL, OpenShift AI REST APIs, LangChain* **Apr.2025 – Jun. 2025**

- Engineered a secure, scalable data infrastructure by migrating the PostgreSQL backend from DigitalOcean to Red Hat OpenShift, integrating IPEDS datasets covering 7,000+ institutions using Python and SQL.
- Collaborated with Red Hat and Akamai teams to provision containerized resources, enhancing data access control, expanding academic coverage, and enabling enterprise-grade deployment scalability.
- Optimized system performance by refactoring complex SQL queries and Python API endpoints on OpenShift, enabling real-time search and accelerating response times, while improving system uptime and reliability.

Traffic Stop Disparity Analysis | R, ggplot2, hypothesis testing, regression analysis ***Jan*. 2025– Apr. 2025**

- Investigated racial and age-based disparities in traffic enforcement outcomes to uncover potential bias in policing and support evidence-based policy decisions.
- Performed statistical testing including one-sample and two-sample t-tests, correlation analysis, and linear regression using R. Created visualizations with ggplot2 (scatter plots, boxplots) to illustrate demographic patterns.
- Identified significant disparities across race and age, revealing that drivers aged 20–40 were disproportionately cited. Findings informed targeted policy recommendations and promoted public awareness.

Time Series Forecasting of Prescription Demand | *ARIMA, Exponential Smoothing, Tableau.* ***Jan*. 2025 – Apr. 2025**

- Addressed the challenge of predicting national prescription demand trends to support proactive healthcare planning and cost control using historical Medicare data.
- Built and compared time series models including ARIMA and exponential smoothing using Python (pandas, statsmodels), and automated trend/outlier detection through rolling windows and decomposition techniques. Designed interactive Tableau dashboards for intuitive data storytelling.
- Produced accurate and interpretable forecasts that uncovered seasonal demand spikes and pricing anomalies, with performance validated through low MAPE and RMSE scores, supporting data-driven healthcare policy decisions and optimized long-term resource planning.